

Monetary policy: Inflation targeting and financial stability pt. 2

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1 Inflation targeting and financial stability

- Two opposing views:
 - Central banks have to reconsider the desirability of (read: abandon) inflation targeting
 - Financial stability is not relevant for monetary policy
- Woodford (2012): inflation targeting needs to be revised
- The CB cares about financial stability and wishes to set the interest rate in such a way that financial imbalances are as small as possible
- The loss function

$$L = \frac{1}{2} [(\pi - \pi^*)^2 + \lambda y^2 + \delta q^2]$$

q is a financial variable measured as a deviation from its mean

- High values of q (think credit) gives “high” probability of a financial crisis / recession
- Low values of q is a sign that it is too hard to get credit
- Woodford (2012): financial crises give loss to society and not only because it leads to a drop in economic activity

1.1 Economic mechanism

- IS and PC equations as before
- The interest rate influences the size of financial imbalances:

$$q = -\phi(r - \rho) + w$$

where w is a shock to financial imbalances

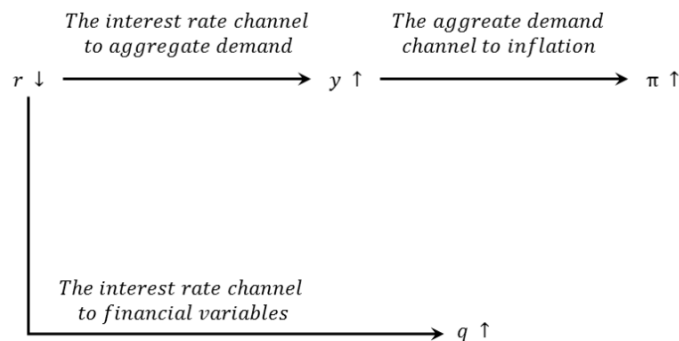
- We can write the equation above as

$$q = -\phi(r - \tilde{r})$$

where $\tilde{r} = \rho + \frac{1}{\phi}w$ is the real rate that closes the financial gap

1.2 The monetary policy transmission mechanism

A reduction in the key policy rate



1.3 Optimal monetary policy

- The central bank minimizes the loss function subject to some equations
- The optimality condition becomes

$$\begin{aligned}
 (\pi - \pi^*) &= -\frac{\lambda}{\gamma}y - \frac{\phi\delta}{\alpha\gamma}q \\
 y &= -\frac{\gamma}{\lambda}(\pi - \pi^*) - \frac{\phi/\alpha}{\lambda/\delta}q
 \end{aligned}$$

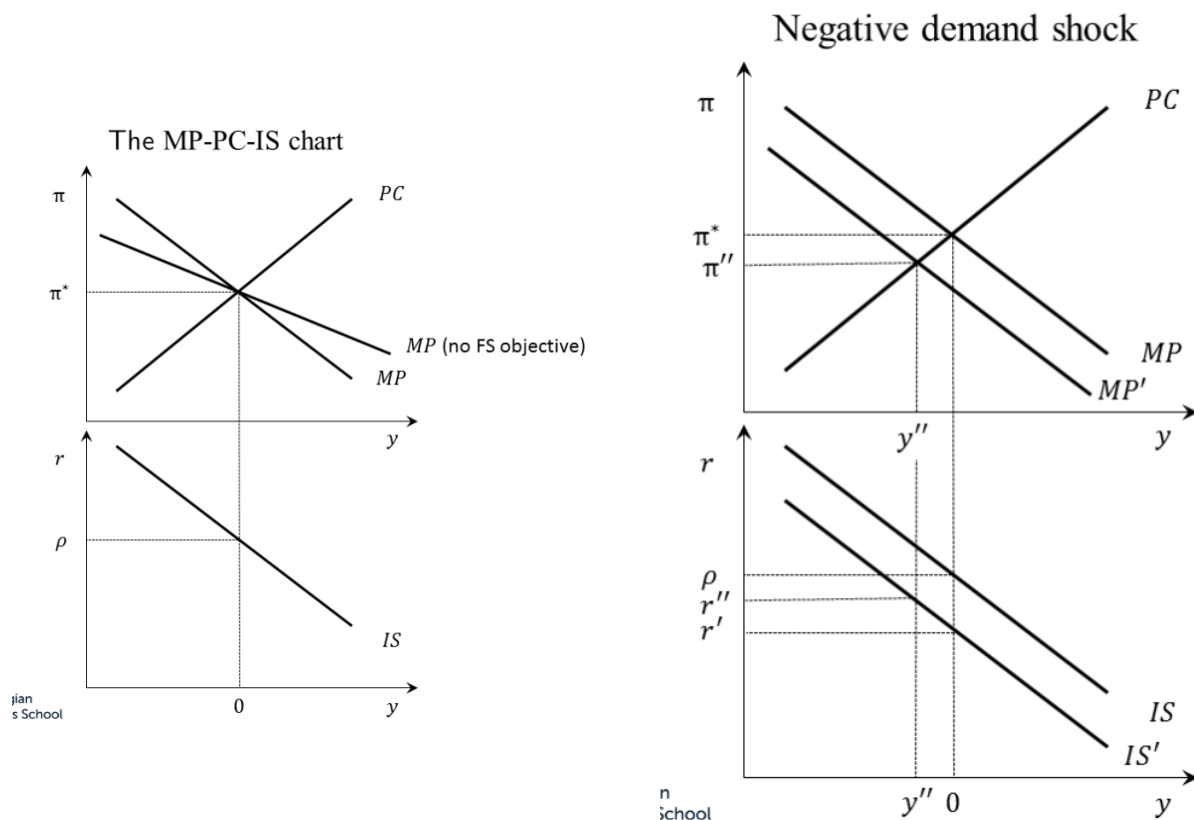
- It can be written as:

$$\begin{aligned}
 (\pi - \pi^*) &= -\frac{\lambda}{\gamma}y - \frac{\phi\delta}{\alpha\gamma}q \\
 &= -\frac{\lambda + \delta\left(\frac{\phi}{\alpha}\right)^2}{\gamma}y - \frac{\phi\delta}{\gamma\alpha^2}(\alpha w - \phi v)
 \end{aligned}$$

1.4 Graphical analysis

- To simplify, we assume $\pi^e = \pi^*$
- We will use two diagrams:
 1. (y, π) -diagram with two equations:
 - The Phillips curve (PC) with positive slope γ
 - Monetary policy (MP) - with negative slope $-\frac{\lambda + \delta\left(\frac{\phi}{\alpha}\right)^2}{\gamma}$
 2. (y, r) -diagram with one equation:
 - IS-equation solved wrt. the real interest rate

$$r = \rho + \frac{1}{\alpha}v - \frac{1}{\alpha}y$$



1.5 A more realistic model

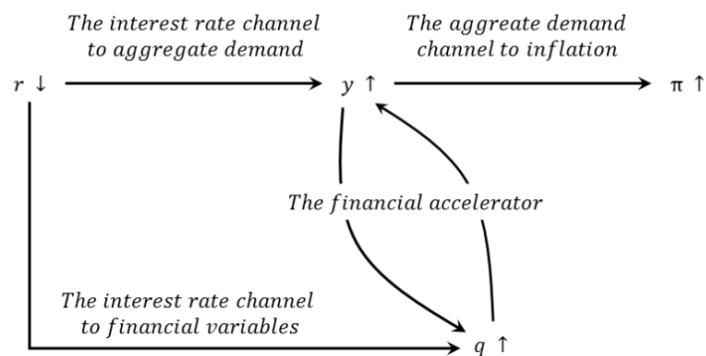
- Financial imbalances tend to build up in booms
- Financial variables feed back to the level of activity and we get a financial accelerator
- More specifically, we assume the following:

$$q = \tau y - \phi(r - \rho) + w$$

$$y = -\alpha(r - \rho) + \chi q + v$$

while the rate of inflation is still given by the old equation

A reduction in the key policy rate



- Our equations can be rewritten as

$$y = -\frac{\alpha + \chi\phi}{1 - \chi\tau}(r - \bar{r}) = -\bar{\alpha}(r - \bar{r})$$

$$q = -\frac{\tau\alpha + \phi}{1 - \chi\tau}(r - \tilde{r}) = -\bar{\phi}(r - \tilde{r})$$

where $\bar{r} = \rho + \frac{1}{\alpha + \chi\tau}v + \frac{\chi}{\alpha + \chi\tau}w$ and $\tilde{r} = \rho + \frac{\tau}{\tau\alpha + \phi}v + \frac{1}{\tau\alpha + \phi}w$

- The optimality condition becomes

$$(\pi - \pi^*) = -\frac{\lambda}{\gamma}y - \frac{\bar{\phi}\delta}{\bar{\alpha}\gamma}q$$